Shrihari Kulkarni

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Education

Vellore Institute of Technology Sep 2021 – Jul 2025
B.Tech in Computer Science & Engineering with Business Systems (CGPA: 8.05) *Vellore, India*

Experience

Software Development Engineer Mar 2026 – Present
Invendis *Bangalore, India*
- Architected and optimized a reporting engine API in .NET/C# across 300+ MySQL tables/views, refactoring query pipelines to achieve a 77% latency reduction (from 6.5s to 1.5s for 1,000+ record queries).
- Engineered dynamic schema-mapping endpoints enabling customizable column aliasing and end-user report composition across 100+ templates with strict multi-tenant isolation.

AI Developer Jul 2025 – Feb 2026
Airrchip *Hybrid*
- Designed and deployed an enterprise RAG pipeline and real-time WebSocket layer delivering cited financial analytics from vector stores with low-latency query reranking.
- Engineered an enterprise LLM Gateway with custom routing algorithms, usage analytics, RBAC, and Redis caching; slashed token and compute overhead by up to 80%.

Data Engineering Intern Jan 2025 – Jun 2025
Medblocks *Bangalore, India*
- Engineered clinical documentation automation pipelines incorporating form auto-population, speech-to-text, and schema validation logic.
- Instrumented Grafana Tempo for distributed tracing across backend microservices to isolate query latency and optimize system throughput.

Projects & Research

FLint: Peer-to-Peer Version Control for DAWs | *C++, JUCE, Node.js* | GitHub Aug 2026
- Built a hybrid VST3 audio plugin using C++ and JUCE, communicating with an asynchronous Node.js sidecar service for robust inter-process communication.
- Implemented decentralized peer-to-peer networking protocols via distributed feeds and swarm discovery to achieve low-latency project synchronization.

MarketMind: Financial Market Analytics RAG Engine | *FastAPI, LanceDB, React* | Website | GitHub Sep 2026
- Architected a lightweight multi-source RAG engine with dynamic time-decay reranking, strictly operating within a <512MB RAM constraint.
- Built a market intelligence platform with React, TypeScript, and Server-Sent Events (SSE) streaming for real-time deep research.

Packaged AutoML Engine | *Python, Scikit-Learn* | PyPI | GitHub Sep 2024
- Engineered an automated model selection package benchmarking linear, tree, and gradient boosting estimators via cross-validation.
- Shipped production-ready PyPI builds featuring dependency isolation, structured entry points, and an intuitive low-code training API.

Regional Ground-Based IoT Solar Irradiance Monitoring | MDPI Sensors Published
- Research publication evaluating solar irradiance across India and Serbia using deep sequence models (TCN-LSTMs, CNN-GRUs, GNNs) trained on NASA POWER IoT telemetry.
- Implemented Modified MissForest and Bidirectional RNN imputation algorithms, significantly outperforming classical baselines under sparse sensor data.

Technical Skills

Developer Tools: Git, Postman, PyTest, Playwright, Claude Code, GeminiCLI
Languages: Python, C#, C++, Java, TypeScript, JavaScript, SQL, GoLang
Backend & Frameworks: .NET Core, ASP.NET, FastAPI, Flask, Spring Boot, REST APIs, Microservices, WebSockets
AI & LLM Systems: RAG, LangChain, LangGraph, Vector DBs (LanceDB, ChromaDB), PyTorch, TensorFlow, Hugging Face
Databases & Cloud: PostgreSQL, MySQL, Redis, AWS, Azure, Docker, Linux, CI/CD, Grafana Tempo
Broad Domain Skills: Machine Learning, Deep Learning, Computer Vision, NLP, Data Engineering, Backend Development, Full Stack Web Development

Certifications

Agentic AI & Product Management (IIT-Patna)
AWS Certified Cloud Practitioner (Amazon Web Services)
Azure Administrator (Microsoft)